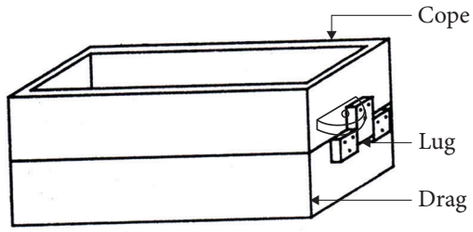
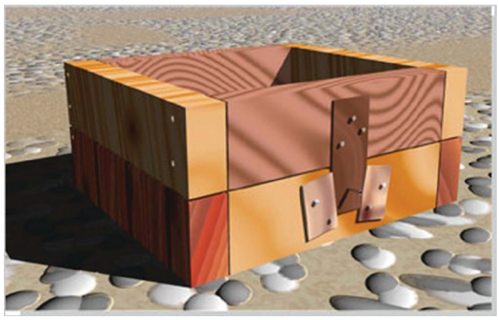
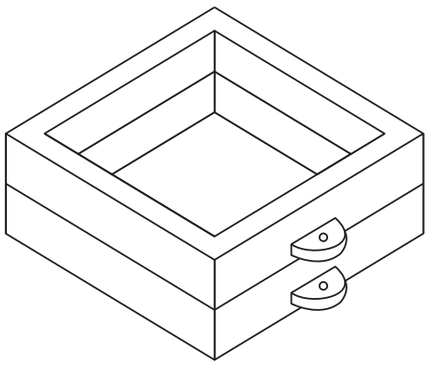
made with hinges and locks to open the flask easily after moulding. Number of moulds can be prepared by using one moulding box.





Snap Flask

Tight or Box Flask: It is used to prepare small and medium size moulds. The box cannot be separated from the mould after moulding. So it is also called permanent flask.





Tight or Box Flask

Moulding: The process of making mould is called moulding. It includes filling of moulding sand around the pattern, ramming, removing the pattern, making runner, riser, gate and vent holes.

Moulding Sand: Moulding sand is an important in foundry. It can withstand the high temperature of molten metal. It does not react with molten metal. It permits the gases and air to escape from the mould when the molten metal is poured. Due to these properties of moulding sand, it is used for casting.

Moulding Sand Ingredients

Moulding sand has the following ingredients

- Sand
- Binder
- Additive

Sand: Sand contains silica, clay and moisture. Sand has 80 to 90% silica which gives refractoriness. Sand contains 5 to 20% clay which gives binding strength. 2 to 3% of water is added with the sand to give moisture.

Binder: Binder is added with the moulding sand to obtain cohesiveness. Binder binds the sand particles together and given strength to the mould. The following three types of binders are used.

Clay Type Binders: Bentonite, Kalvanite are clay type binders.

Organic Binders: Wood, resin, linseed oil dextrin and molasses are organic binder.

Inorganic Binders: Portland cement and sodium silicate are inorganic binders.

Additives: Additives are added with the moulding sand to improve the

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